

Prompt Engineering & Experimentation

Date	2 October 2026
Team ID	SWTID-2026-6582
Project Name	PocketSmart AI: Your Smart Budget & Recommendation Assistant
Maximum Marks	Project Deliverable

Initial Prompt Design

Initial prompts supplied planner inputs and asked Gemini to return relevant recommendations within the user budget.

System Instructions / Role Definition

Prompts frame Gemini as a budget-aware recommendation assistant for the selected domain and require structured, context-sensitive output rather than generic prose.

Zero-shot / Structured Prompting

The implementation primarily uses structured zero-shot prompting: user requirements are inserted into domain-specific prompt templates and Gemini is instructed to return data matching the expected schema.

Prompt Testing Iterations

- Compared different party requirements at the same budget to ensure allocations change with context.
- Tested Home and Jewelry text generation through the Gemini service.
- Tested Jewelry with an actual image object to verify multimodal invocation.
- Tested over-budget plans to trigger a revision request instead of silently accepting them.

Output Evaluation Criteria

- Relevance to user inputs.
- Structured response parsable by Pydantic.
- Budget feasibility after Python recalculation.
- Meaningful variation across materially different scenarios.
- No fake static fallback when Gemini fails.

Prompt Optimization

Prompts request item names, categories, quantities, estimated unit prices, descriptions/reasoning and search terms. Revision prompts include the target budget and calculated excess so Gemini can rebalance the plan.

Safety & Bias Handling

The application validates inputs and treats AI output as recommendations rather than guaranteed facts. Estimated prices are not represented as live prices. Financial arithmetic is recalculated independently in Python.

Final Optimized Prompt Pattern

ROLE + USER CONTEXT + DOMAIN REQUIREMENTS + BUDGET CONSTRAINT + REQUIRED STRUCTURED FIELDS + ESTIMATED-PRICE DISCLOSURE + OUTPUT SCHEMA. For Jewelry, the validated outfit image is supplied together with the text prompt.